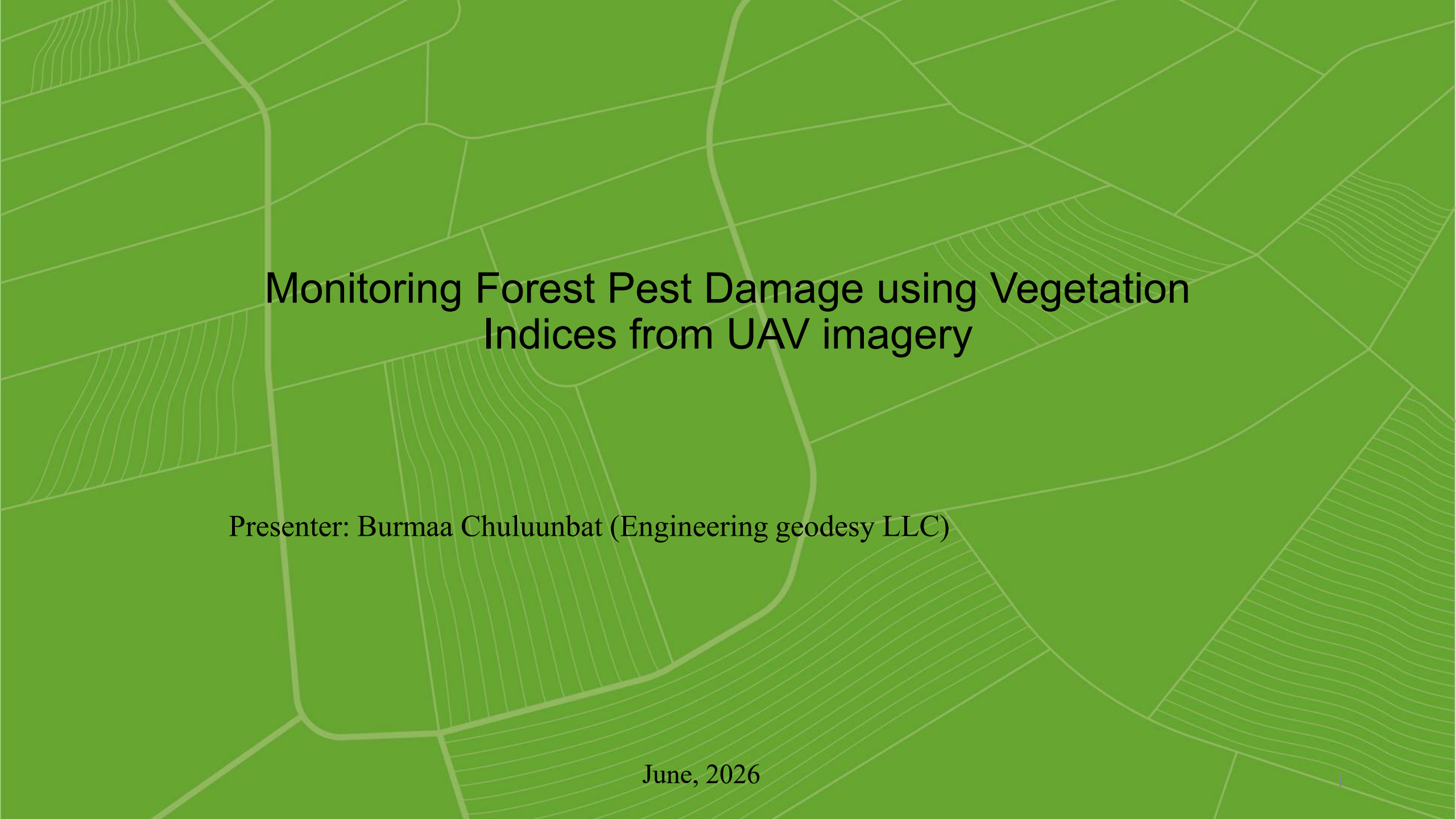


The background of the slide is a green topographic map. It features a network of thin white lines representing contour lines and a thicker white line representing a road or path. The contour lines are more densely packed in some areas, indicating steeper slopes, and more spread out in others, indicating flatter terrain. The overall pattern is a complex web of lines that define the geographical features of the area.

Monitoring Forest Pest Damage using Vegetation Indices from UAV imagery

Presenter: Burmaa Chuluunbat (Engineering geodesy LLC)

June, 2026

CONTENTS

1. Introduction

- Research intro
- Forest pest impact
- Research area
- History

2. Methodology

- UAV flight operations
- Pest identification

3. Processing

- LiDAR data
- MS data

4. Results

- Analysis
- Recovery

INTRODUCTION

The Challenge

The spread of forest pest insects has notably increased in recent years, negatively impacting forest regeneration and growth in Mongolia. The Siberian silk moth (*Dendrolimus sibiricus*) has caused considerable damage.

Current method limitations



The Solution

UAV technology enables efficient forest pest monitoring in remote regions.

Target Pest

Dendrolimus sibiricus
Siberian silk moth

LOCATION

Gorkhi-Terelj National Park,
Mongolia

Tree type

Larch forest



Standard Forest Pest Control Costs by Method



Source: Forest Agency, Implementing Agency of the Government of Mongolia

RESEARCH AREA & PURPOSE

Objectives: To develop and evaluate an integrated UAV-based monitoring and treatment assessment framework for detecting forest pest infestation, applying biological control, and quantifying forest recovery using multispectral and LiDAR data.



Forest Pest Outbreak Pattern

Siberian Silk Moth *Dendrolimus sibiricus*

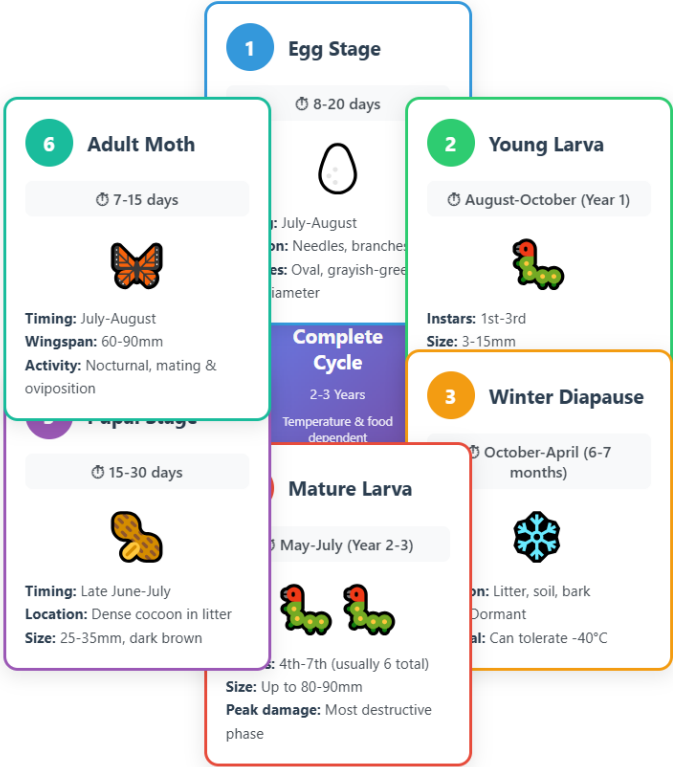
Major defoliator of larch forests. An outbreak has been detected in the selected study site, with severe infestation concentrated in the central forest stand.

Infestation Factors (In Research area -Central Stand)

- Microclimate
- Higher Humidity
- Stagnant Air

Siberian Silk Moth Life Cycle

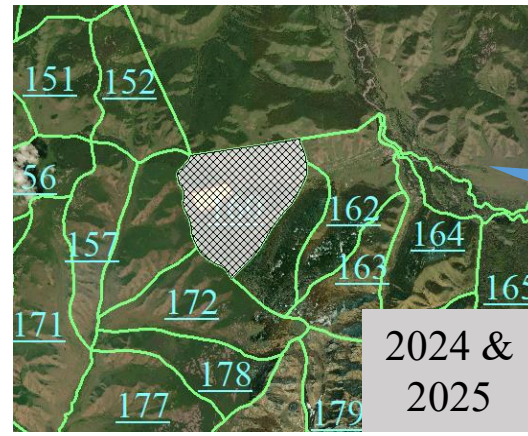
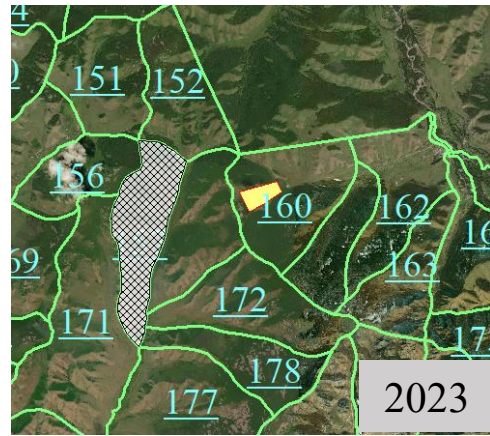
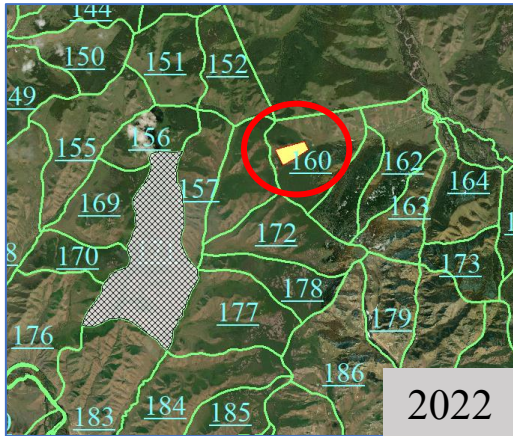
Dendrolimus sibiricus Tschetverikov - Complete Development Cycle



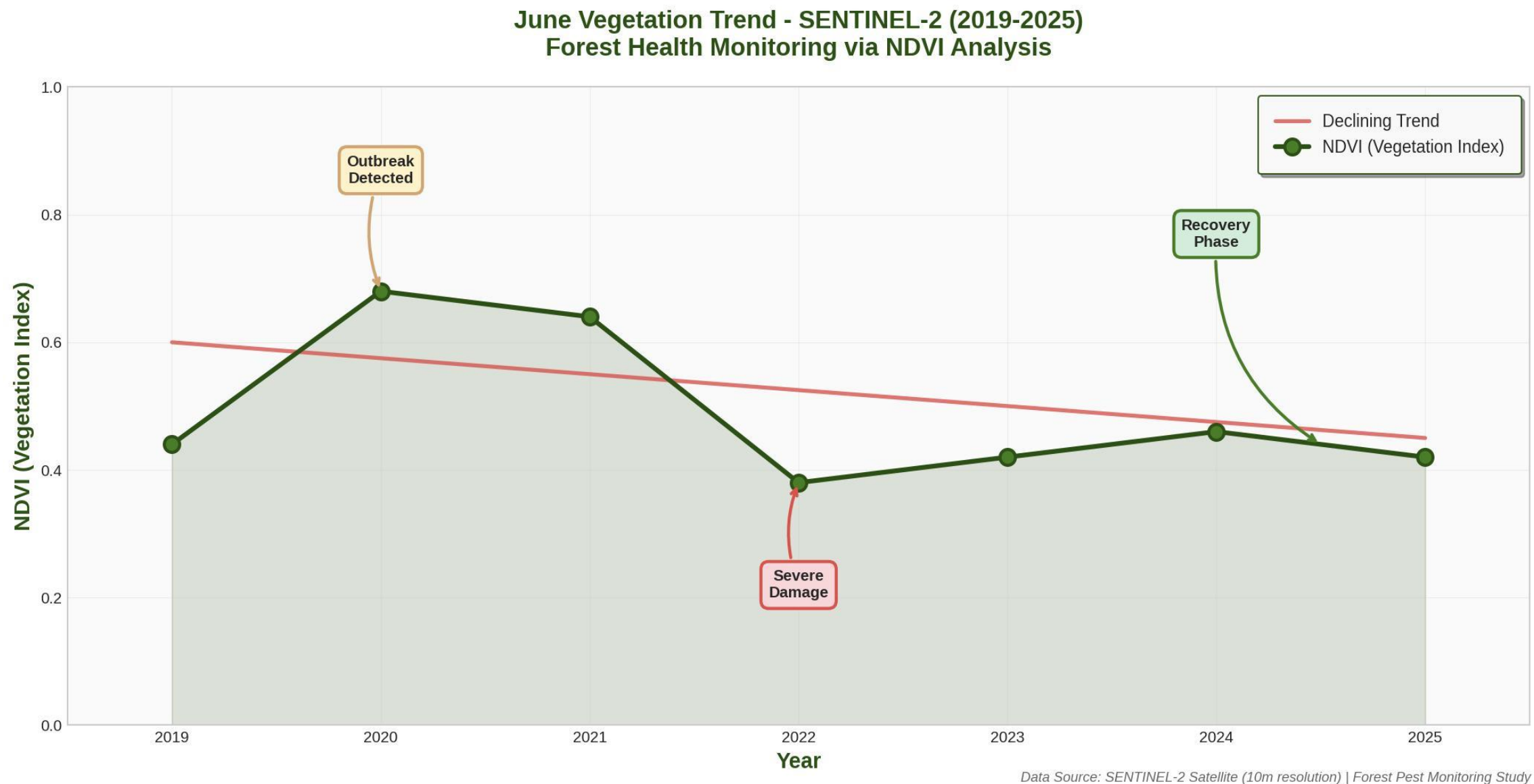
Caterpillar stage

HISTORY OF PEST DAMAGES:

- Field research work based on the findings of the forest pest outbreak between 2022-2025



• 7-Year Vegetation Investigation: SENTINEL-2 NDVI Analysis



INSECT PEST MANAGEMENT

Combat with pest by airplane

Bacillus thuringiensis
var. kurstaki



June 4, 2025

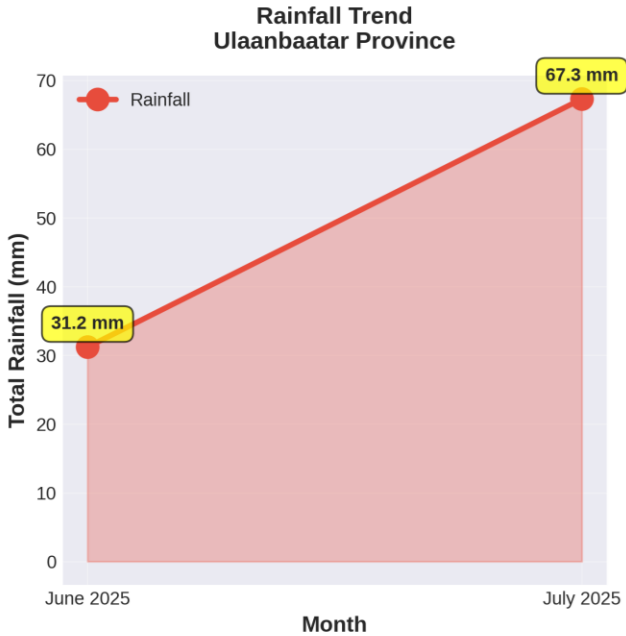
Aircraft JU-1918 for spray



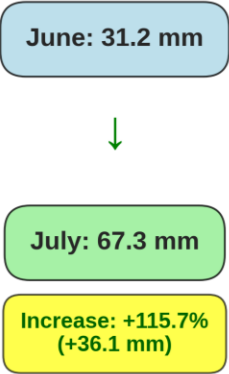
Biological control agent



Spraying work in Terej
June 4, 2025



Month-to-Month Change Analysis



RAINFALL SUMMARY	
Nalaikh District, Ulaanbaatar	
Station Code: 480075	
Total Rainfall: 98.5 mm	
Average: 49.2 mm	
Wettest Month:	
July 2025 (67.3 mm)	
Driest Month:	
June 2025 (31.2 mm)	
Data Source:	
National Meteorology Agency	

METHODOLOGY

I. Data collection

Multispectral
images

LiDAR point cloud



II. Data processing



DJI TERRA

DJI TERRA

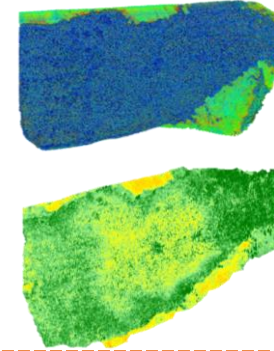
Python – Point Cloud

Google Earth Engine
– Vegetation indices

DJI MODIFY – Point
Cloud



Google Earth Engine



III. Insect pest management

Aircraft spray on
the detected
hotspot

Aircraft JU-1918
for spray

Drone spray on the
detected hotspot

DJI Agras T50 for
spray (2026)



Bacillus thuringiensis
var. kurstaki



IV. Results

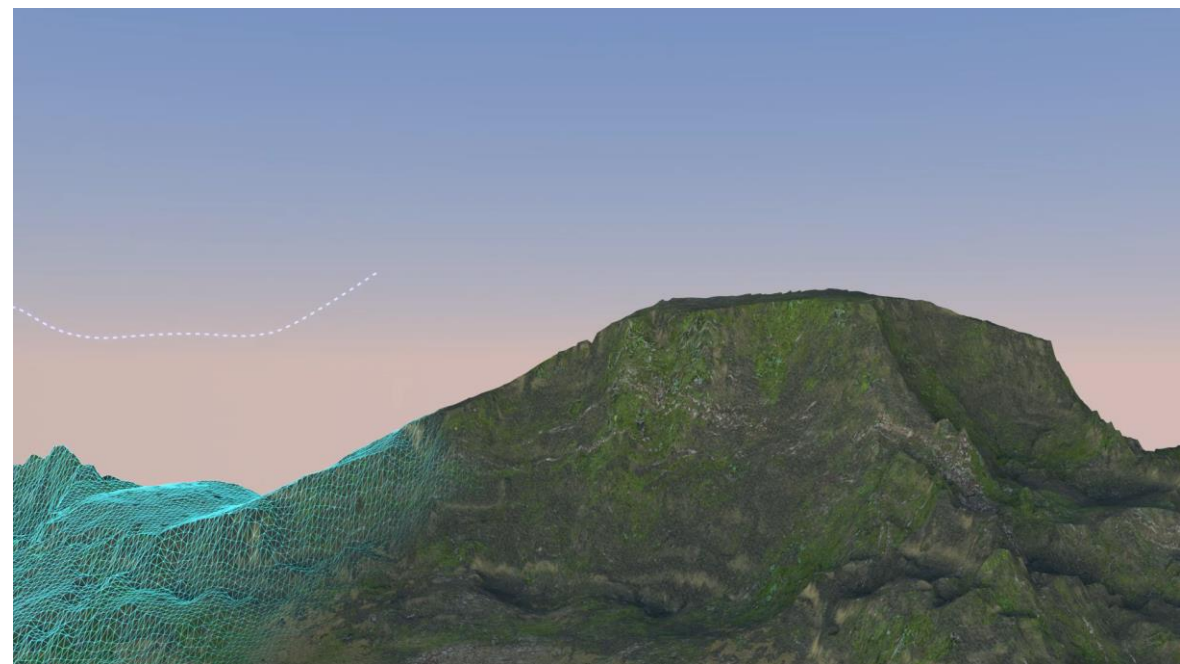
- ANALYSIS
- METHOD COMPARISON (Forest recovery)
- DISCUSSION

Data acquisition



M3M sensor specification

Drone	Band	Wavelength
DJI Mavic 3 Multispectral RGB camera 20MP	Green	560±16nm
	Red	650±16nm
	Red Edge	730±16nm
	Near Infrared	860±26nm



UAV LiDAR Data



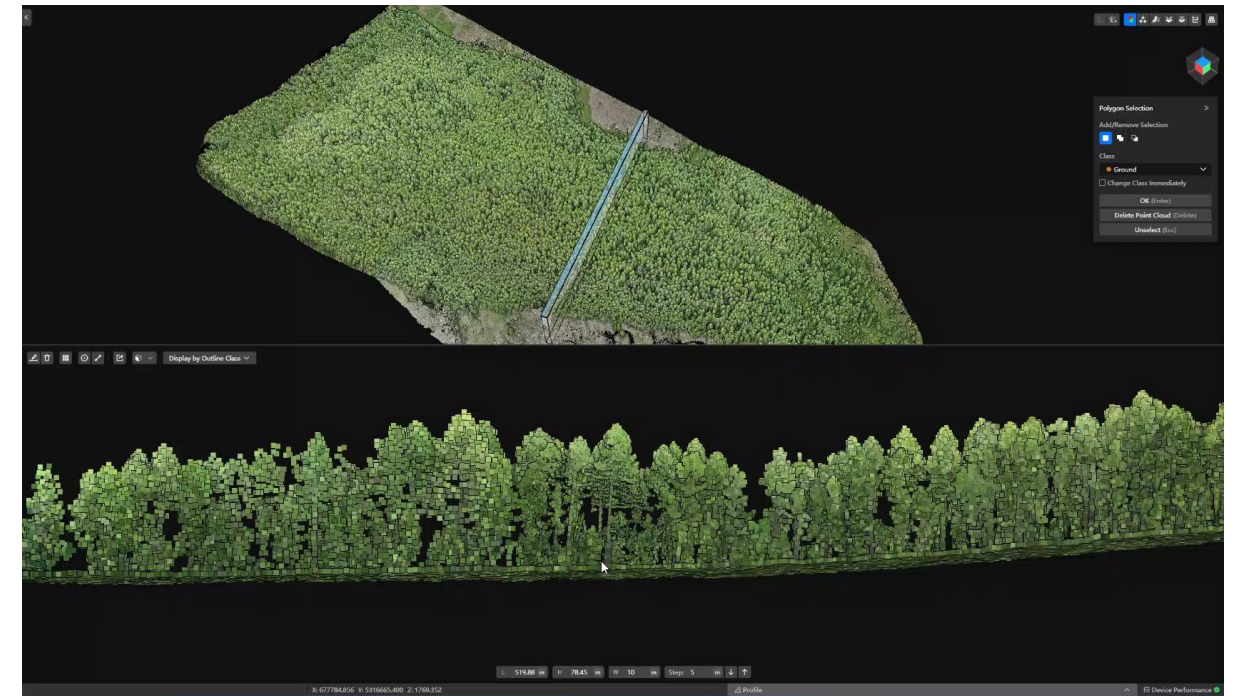
PERFORMANCE PARAMETERS:

Field of view	70°
Laser wavelength	905nm
Laser spot size	4sm horizontal 12sm vertical
Scan mode	Non-Repetitive
Weight	905±5 g

Forest Structure Comparison Using LiDAR Point Clouds (3rd of June and 18th of July)



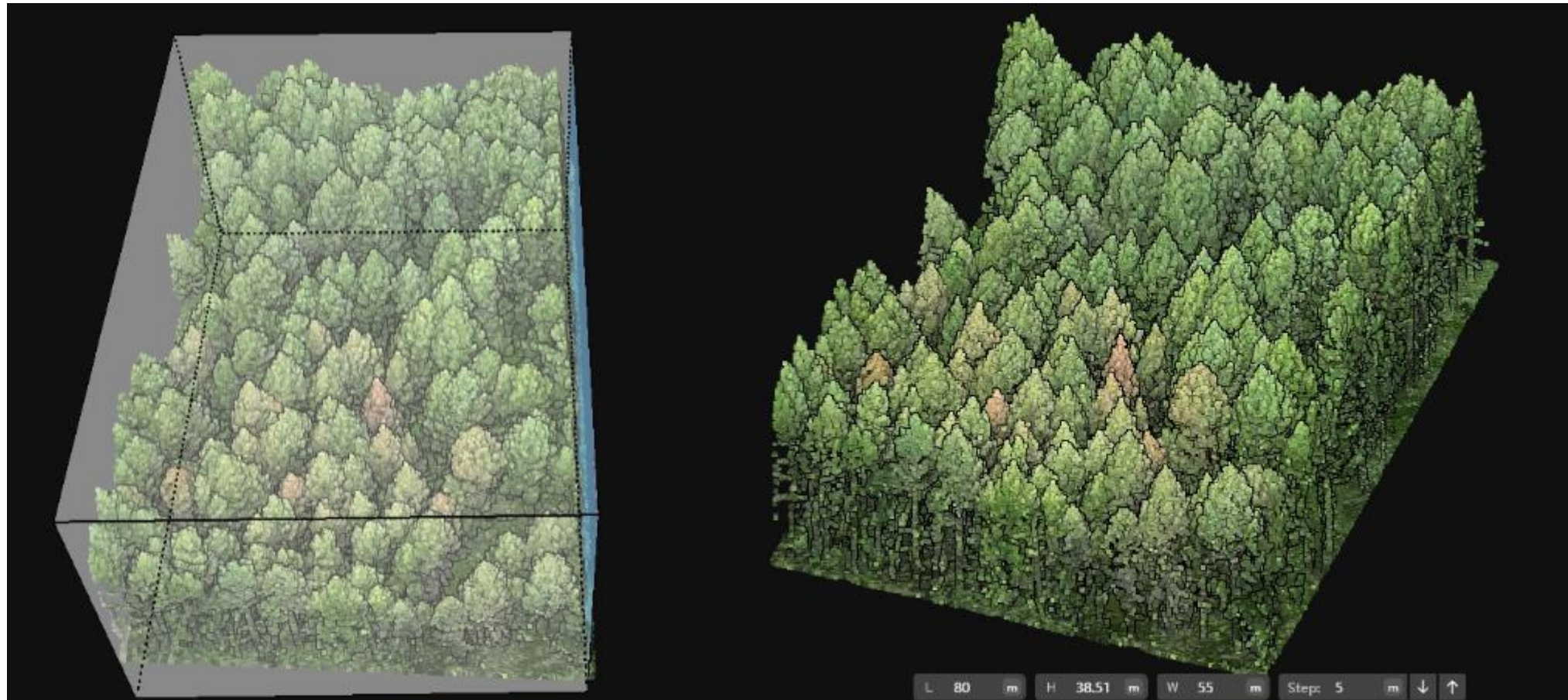
Date: 3rd of June

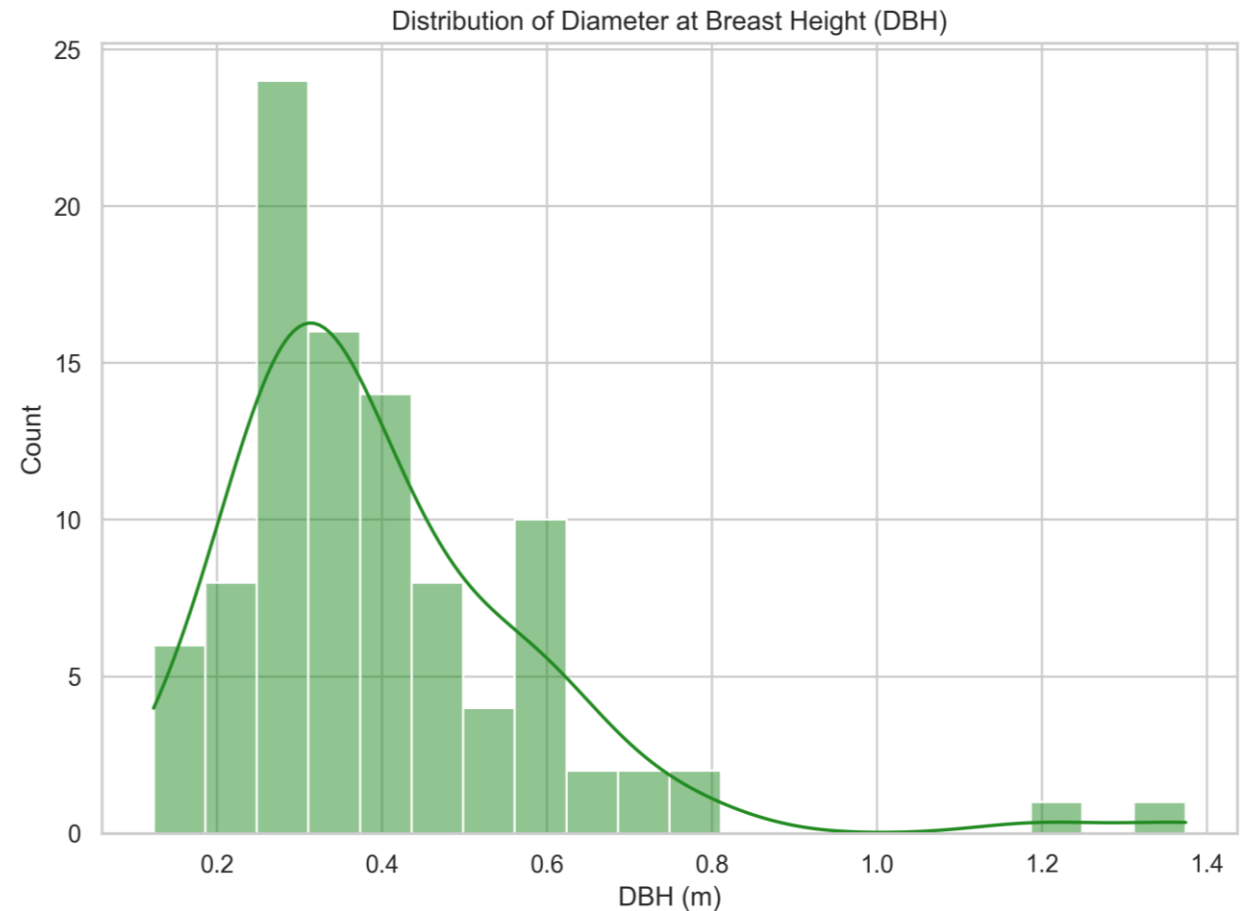
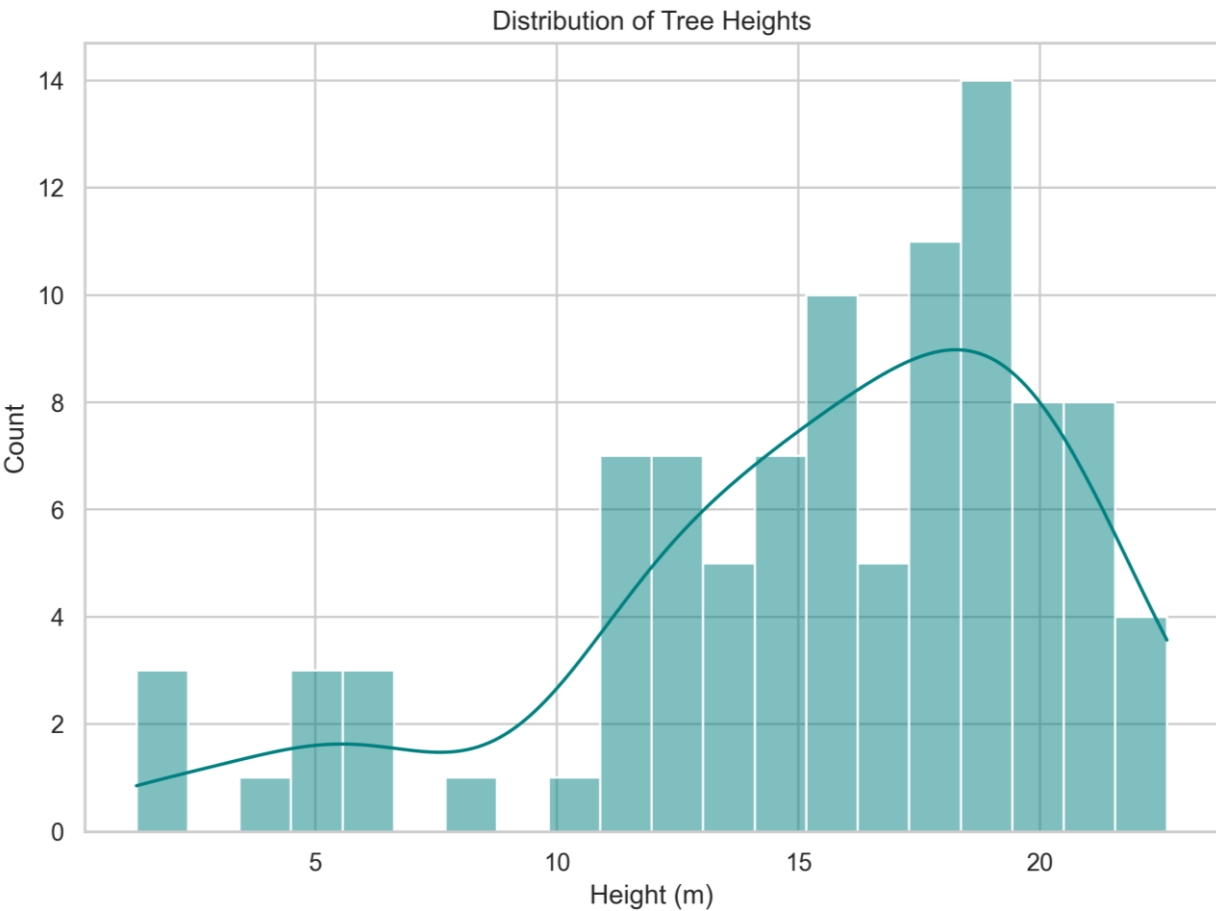


Date: 18th of July

Software: DJI Modify

Forest Tree Inventory – Lidar data (Test area – affected zone 55m x 80m)





Data date: 18th of July, 2025

Algorithm: The Cloth Simulation Filter (CSF)

Leaf Area Index (LAI) from LiDAR point clouds

No	Vegetation index	Formula	References
1	Leaf Area Index (LAI)	$NDVI = (NIR - RED) / (NIR + RED)$	Chen, J.M. & Black, T.A. (1992)

Leaf Area Index (LAI) created from LiDAR data and using 3 methods:

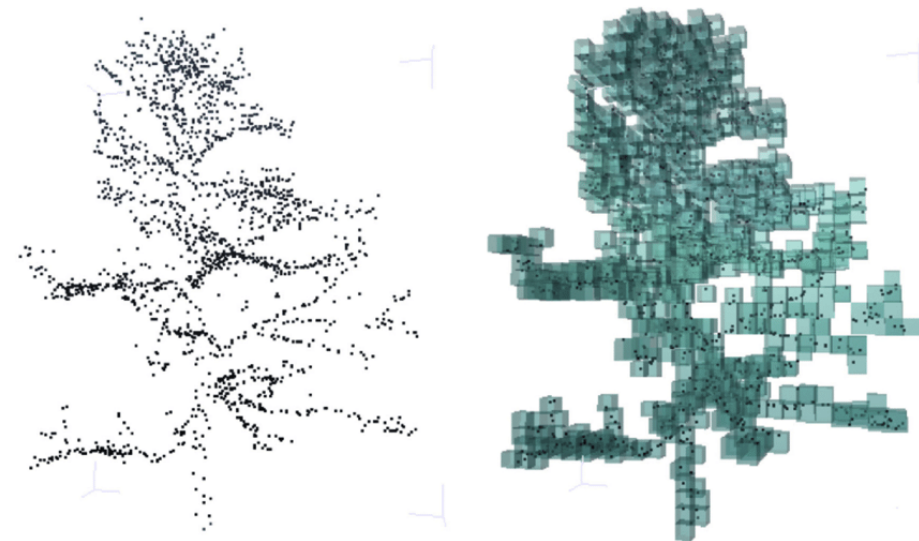
Three-Method Approach: Voxel (50%) + Point Density (30%) + Height-Weighted (20%)

Equation:

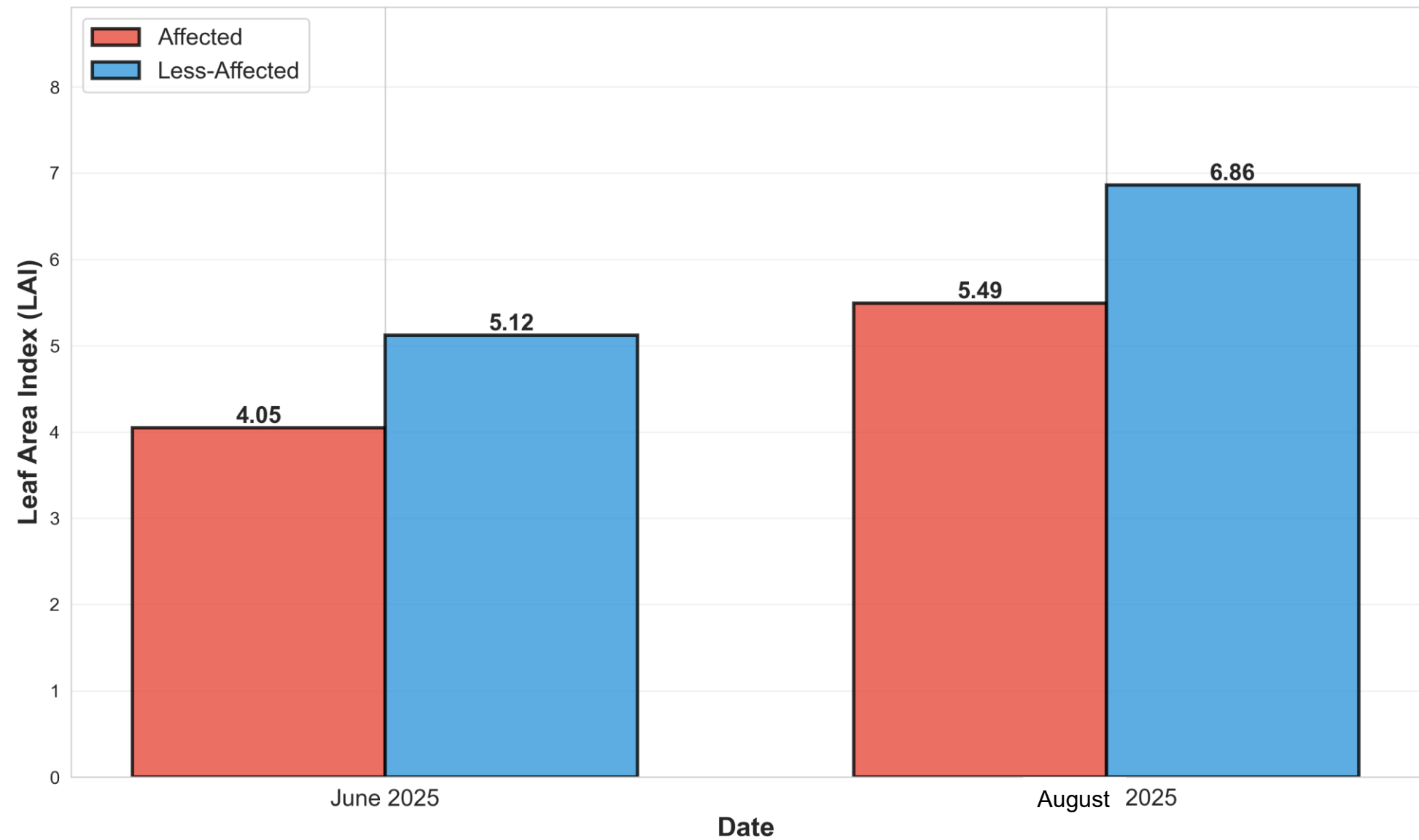
- $i=[dx-xmin], j=[dy-ymin], k=[dz-zmin]$
- $$V(i,j,k) = \begin{cases} 1 & \text{if } \geq \text{LiDAR point inside} \\ 0 & \text{if empty} \end{cases}$$
- $$LAI = -\ln \left(\frac{N_{empty}}{N_{total}} \right)$$

A **voxel** (volumetric pixel) is a small 3D cube, like a **pixel** but in **3D**.

- Pixel → 2D square
- Voxel → 3D cube



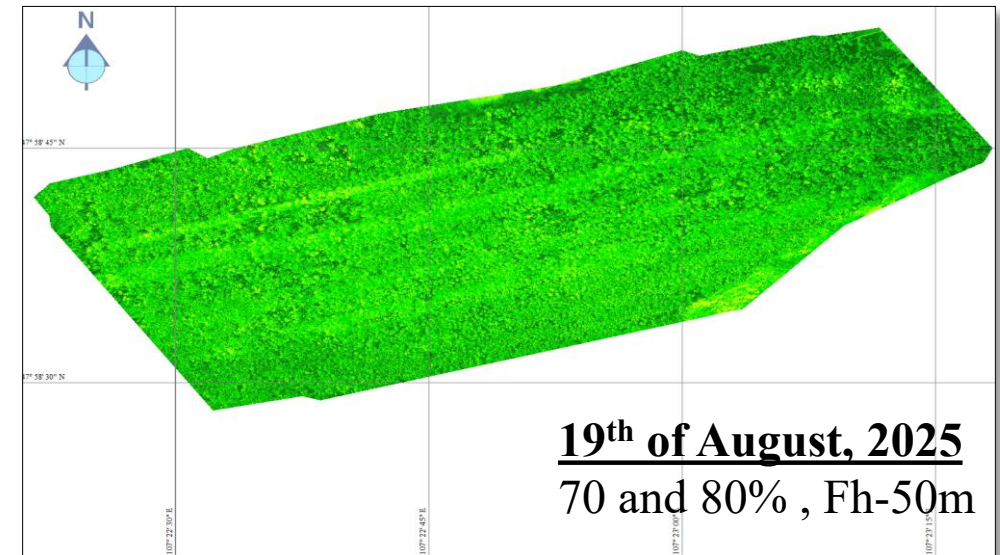
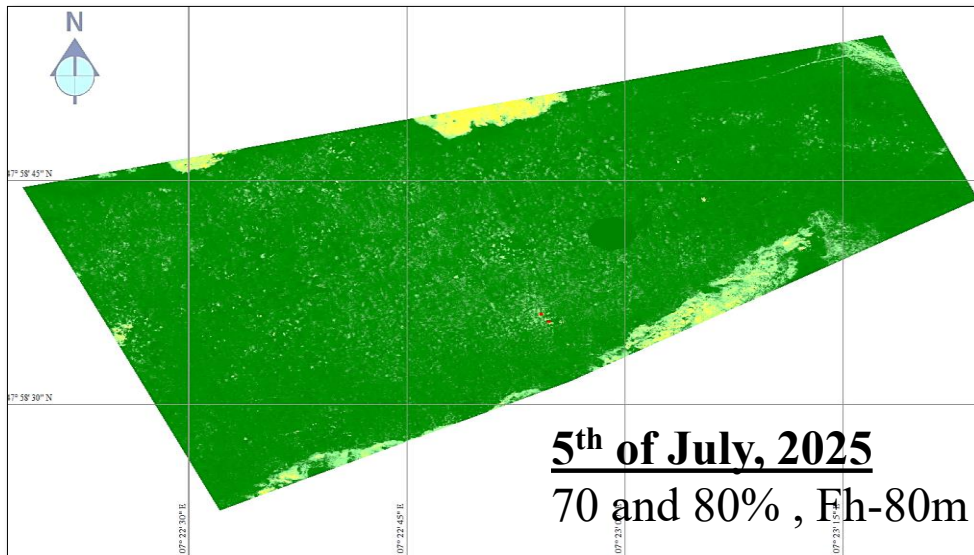
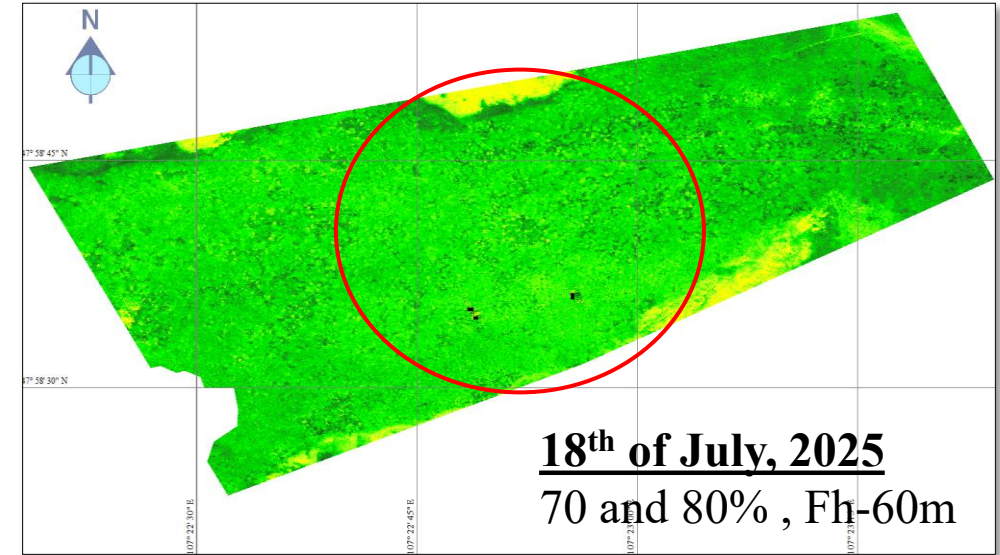
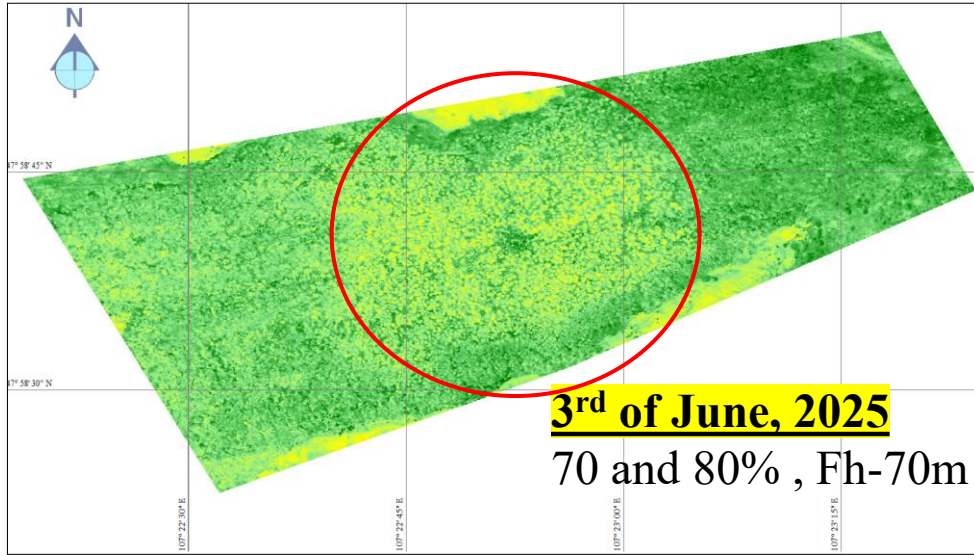
LiDAR-Derived LAI Comparison
Gorkhi-Terelj National Park



Multispectral vegetation index calculation

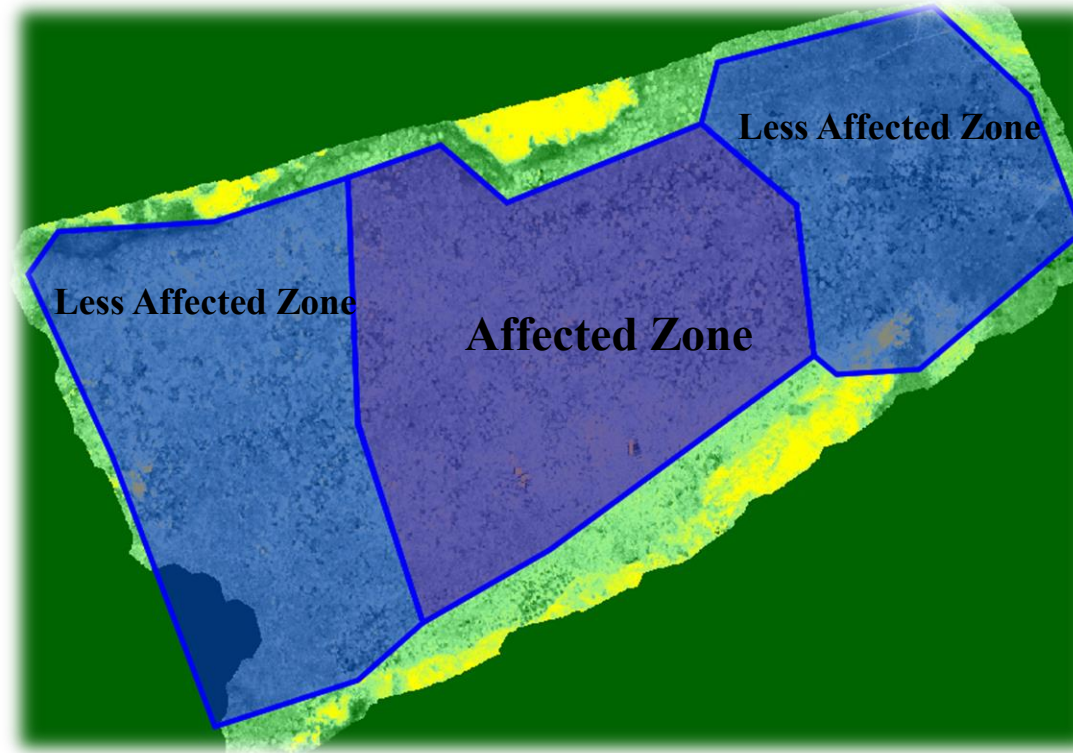
№	Vegetation index	Formula	References
1	Normalized Difference Vegetation Index (NDVI)	$NDVI = (NIR - RED) / (NIR + RED)$	Rouse, J.W., Haas, R.H., Schell, J.A., & Deering, D.W. (1973)
2	Green Normalized Difference Vegetation Index (GNDVI)	$GNDVI = (NIR - GREEN) / (NIR + GREEN)$	Gitelson, A.A., Kaufman, Y.J., & Merzlyak, M.N. (1996)

Data collection: GNDVI images between June and August

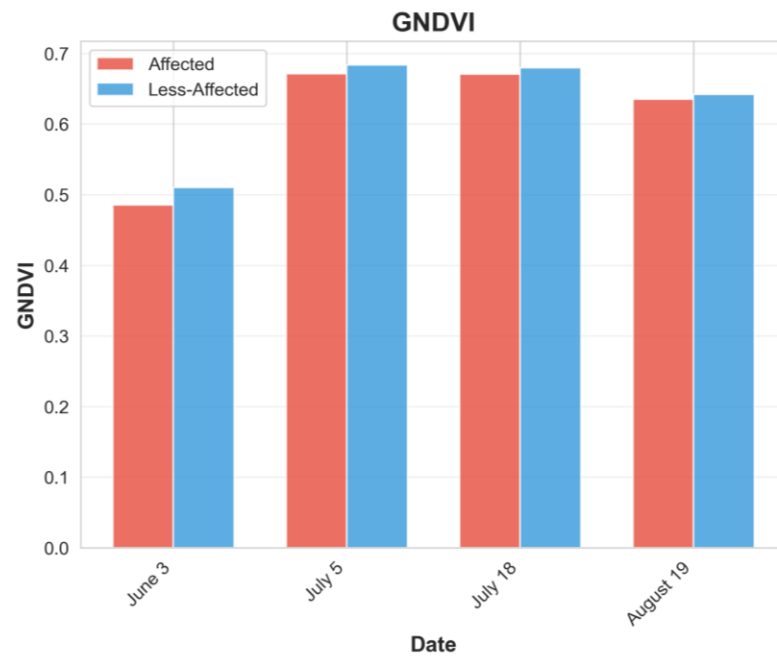
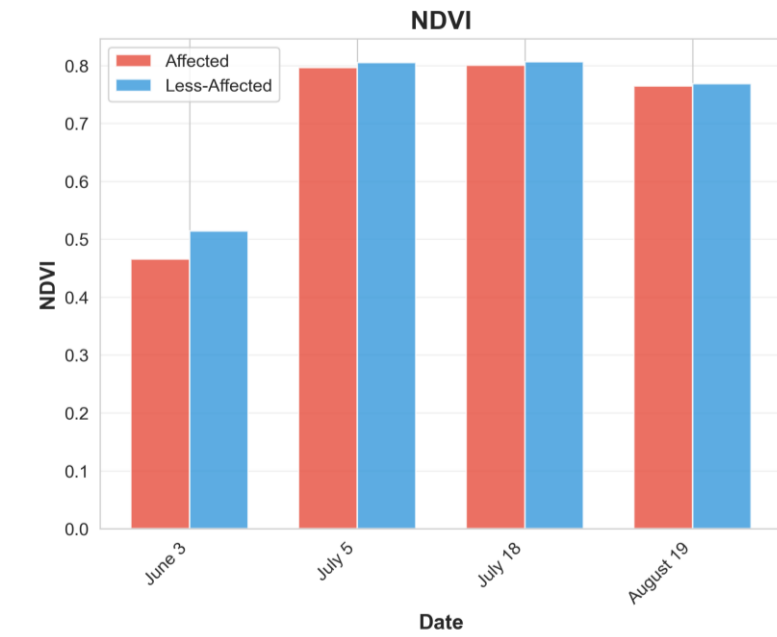


Area Classification for Pest Control Strategy

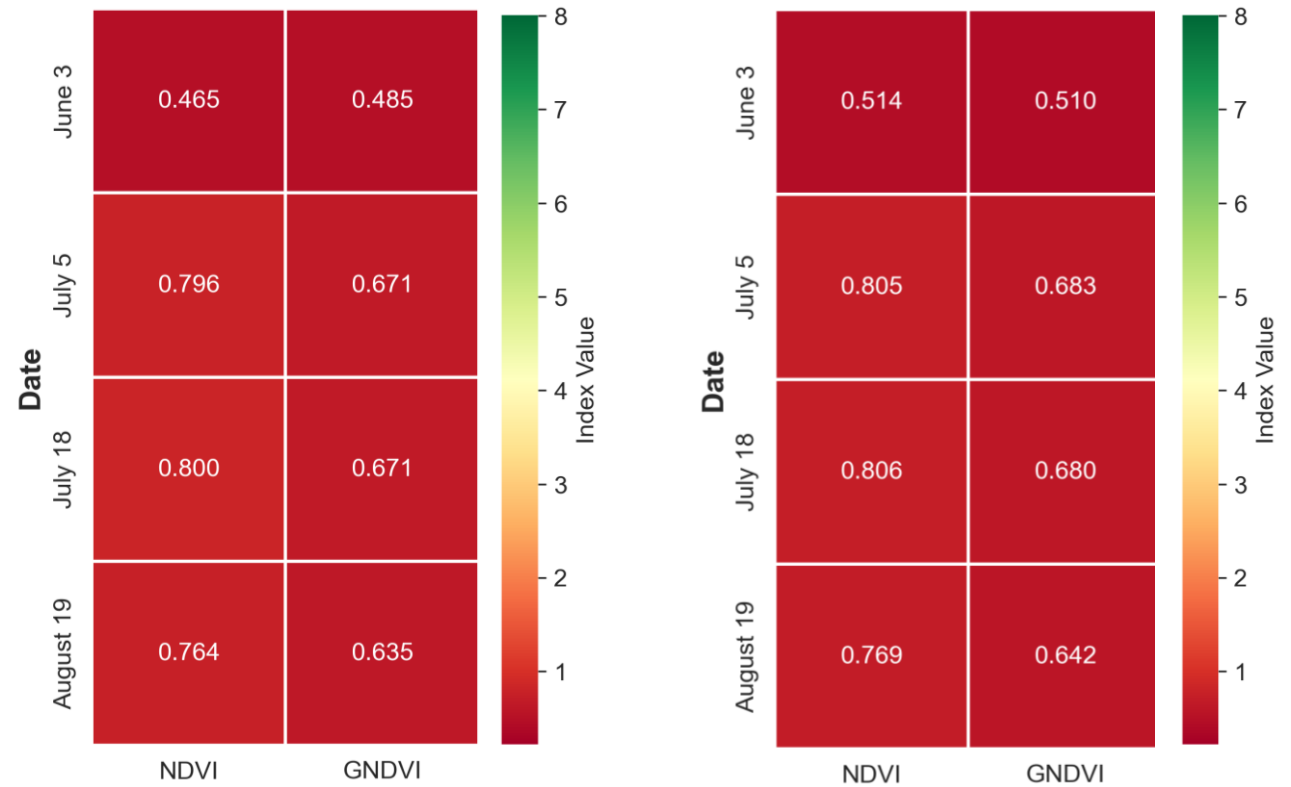
- **Affected Zone**
 - High pest density observed
 - Visible canopy damage
- **Less Affected Zone**
 - Lower pest presence
 - Minimal canopy impact



MULTISPECTRAL DATA ANALYSIS : Index Comparison by Date

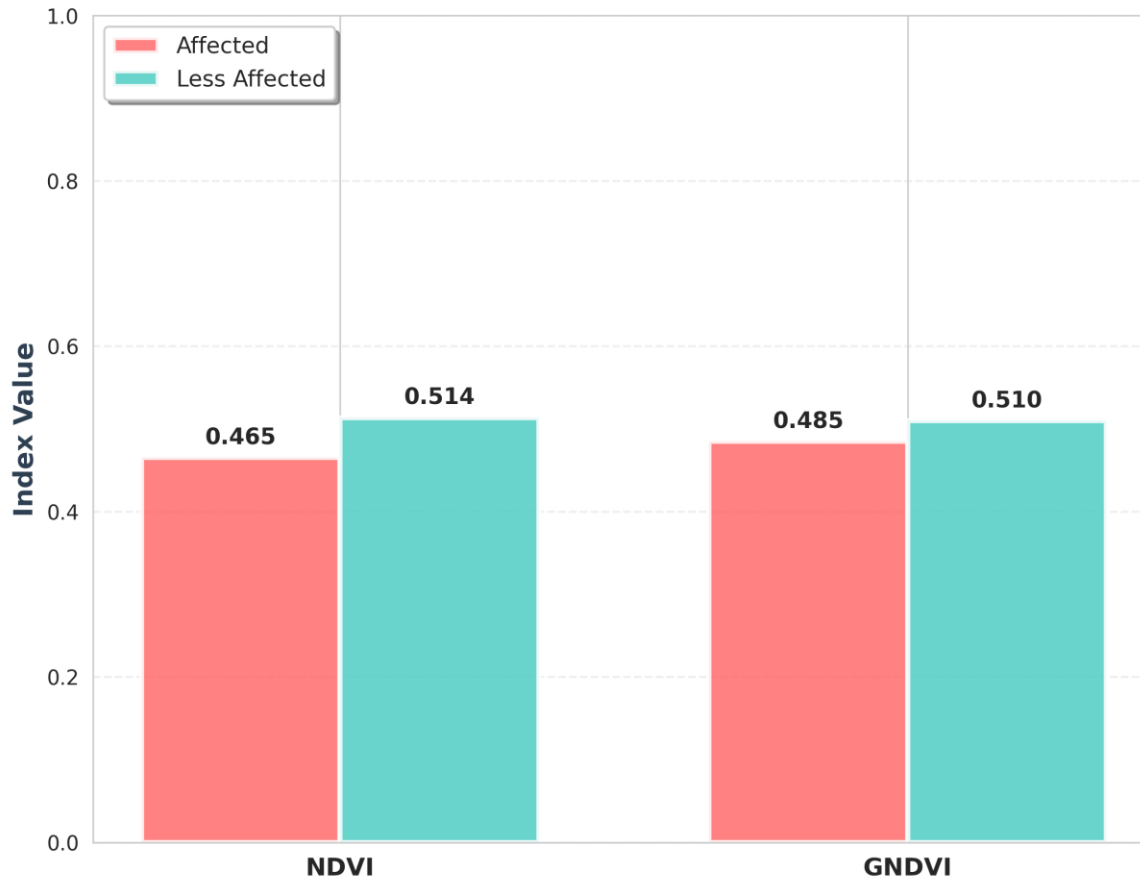


Index Values Heatmap: Date x Index

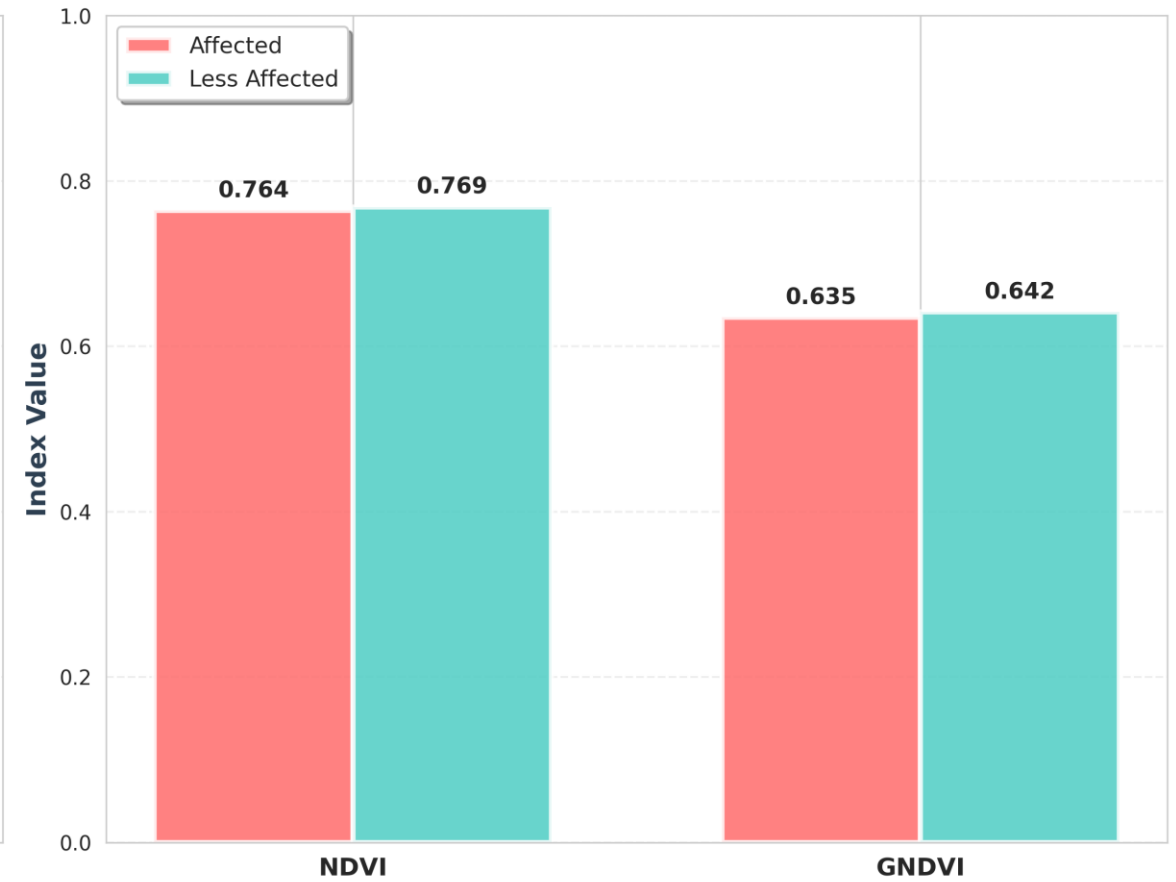


Vegetation Index Comparison: Affected vs Less Affected Areas

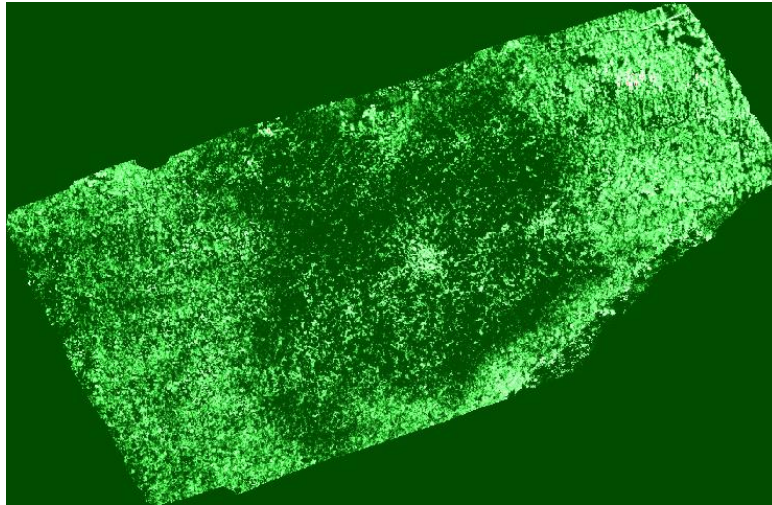
June 3, 2025



August 19, 2025



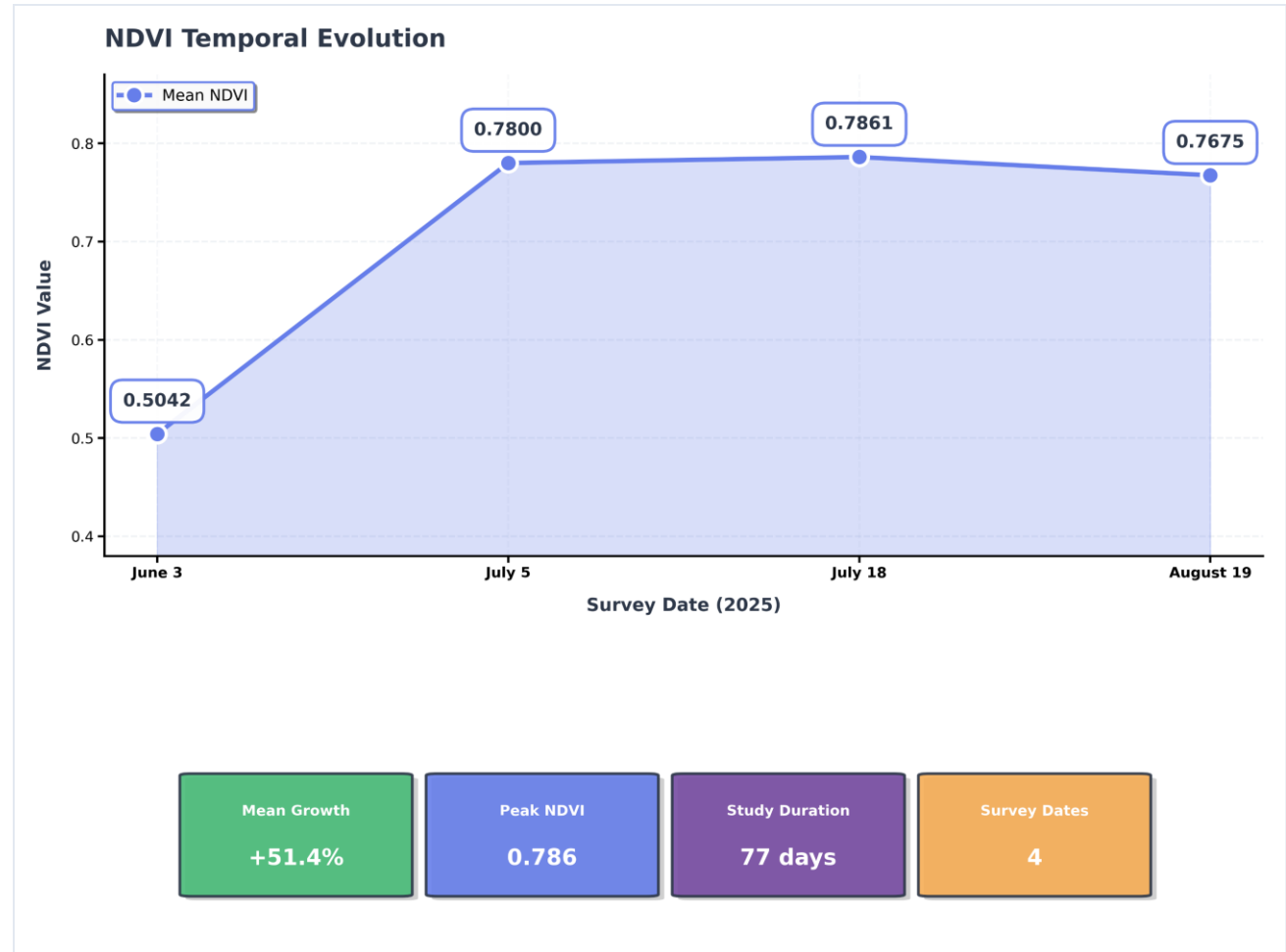
Conclusion



NDVI Change Categories

June → August 2025

-  Moderate Increase (0.05 to 0.1)
-  Strong Increase (0.1 to 0.15)
-  Very Strong Increase (> 0.15)



References

- [1] Wang, Y.-H.; Zhou, G.-S.; Jing, Y.-L.; Yang, Z.-Y. Estimating Biomass and NPP of Larix Forests using forest inventory data. *Chin. J. Plant Ecol.* **2001**;
- [2] He-Ya Sa.;Xiaojun Huang.; Li Ling.; Debao Zhou.; Junsheng Zhang 4.; Gang Bao.; Siqin Tong.; Yuhai Bao.; Dashzebeg Ganbat.; Mungunkhuyag Ariunaa.; Dorjsuren Altanchimeg .; Davaadorj Enkhnasan. Multi-Dimensional Estimation of Leaf Loss Rate from Larch Caterpillar Under Insect Pest Stress Using UAV-Based Multi-Source Remote Sensing. *Drones.* 2025;
- [3] Ting Zhang.; Yameng Yu.; Yupu Lu.; Hao Tang.; Kai Chen.; Jiahui Shi.; Zeqi Ren.; Shuilin Wu.; Dandan Xia.; Yufeng Zheng a Integrating UAV LiDAR and multispectral data to assess forest status and map disturbance severity in a West African forest patch;
- [4] Morsdorf et al. (2006) - Forest Ecology & Management"Assessment of the influence of flying altitude and scan angle on biophysical vegetation products derived from airborne laser scanning"Point density methods;
- [5] Zhao & Popescu (2009) - Remote Sensing of Environment"Lidar-based mapping of leaf area index and its use for validating GLOBCARBON satellite LAI product"Multi-method integration;
- [6] Richardson et al. (2009) - Agricultural & Forest Meteorology"Distinguishing vegetation from soil background information"UAV-LiDAR specific;
- [7] Wallace et al. (2012) - Remote Sensing"Assessment of Forest Structure Using Two UAV Techniques“.

Thank you for your attention

